

Juan Azcue Estrada

Spanish citizen | 23 years old

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Bonn, Germany

RESEARCH INTERESTS & BACKGROUND

I am a graduate student knowledgeable in differential geometry (fibre bundle theory, Lie groups, symplectic geometry, Kähler geometry, see the courses below) and sheaf theory, currently learning about Floer theory and Mirror Symmetry. I have a keen interest in the interactions between representation theory and several flavours of geometry (particularly complex/ algebraic) which often lie at the interface with mathematical physics.

EDUCATION

Rhenish Friedrich Wilhelm University of Bonn

Oct 2024 – Present

Master of Science in Mathematics (Average Grade: 1.7)

- **Representation and structure theory of compact semisimple Lie groups:** Prof. Dr. Tasho Kaletha (9 EC).
- **Algebraic Geometry II:** Prof. Dr. Daniel Huybrechts (9 EC).
- **Symplectic Geometry (Selected Topics):** Prof. Dr. Laurent Côté and Dr. Nathaniel Bottman.
- **Moduli spaces of vector bundles over curves:** Prof. Dr. Huybrechts and Dr. Carini (5 EC).
- **Complex Analysis (Several variables & Sheaf theory):** Prof. Dr. Ingo Lieb (7 EC).
- **Constructive Quantum Field Theory:** Prof. Dr. Christian Brennecke (7 EC).
- **Topology I (Homotopy & Homology):** Prof. Dr. Stefan Schwede (9 EC).
- **Three-dimensional manifolds:** Dr. Paula Truöl (5 EC).

University of the Basque Country (UPV/EHU)

Sep 2020 – Jun 2024

Bachelor of Science in Mathematics (Grade: 9.35 / 10.0)

- **Extraordinary Graduation Prize:** Awarded to the top 2 students out of 80.
- **Bachelor Thesis:** “Critical points in symplectic topology”. Supervised by Dr. Álvaro del Pino Gómez. Investigation of Morse theory and the Arnold conjecture using elementary (non-Floer) techniques (80 pages).

Erasmus Program at Utrecht University (Mastermath National Programme)

Sep 2023 – Jun 2024

- **Symplectic Geometry:** Prof. Gil Cavalcanti and Prof. Federica Pasquotto (8 EC).
- **Differential Geometry:** Prof. Marius Crainic (8 EC).
- **Algebraic Geometry 1 & 2:** Profs. M. Kool, M. Hablicsek, L. Taelman, and N. Olander (16 EC).
- **Elliptic Curves:** Prof. Emma Brakkee and Prof. Valentijn Karemaker (8 EC).
- **Algebraic Number Theory:** Prof. Peter Stevenhagen and Jan Bonk (8 EC).

RESEARCH EXPERIENCE & SUMMER SCHOOLS

- **ICMAT Research Scholarship:** “The Calderón problem and pseudo-holomorphic curves” supervised by Dr. Daniel Faraco Hurtado (Jun – Aug 2023).
- **BCAM Courses:** Symplectic Geometry (Dr. Marisa Fernández) and Multiplicity Theory (Dr. Ilya Smirnov).
- **JAIE Summer School (ICMAT):** Courses by P. Tradacete (Banach lattices), M.A. García Ferrero (Inverse problems), A. Ibort (Game theory), T.L. Gómez de Quiroga (Algebraic Geometry).
- **ALGANT Summer School 2025:** “Algebraic groups and algebras with involution”.

TEACHING EXPERIENCE

- **Tutor (University of Bonn):** Exercise lessons for "Foundations in Representation Theory" (Complex semisimple Lie algebras) for Prof. Dr. Kaletha (Winter 2025-2026).

LANGUAGES & SKILLS

- **Languages:** Spanish (Native), English (Advanced/C1).
- **Technical:** LaTeX, Scientific Computing (Python/Matlab/Mathematica).